Deepfake Detection with Adversarial Robustness

1. Introduction

Deepfake technology has rapidly advanced with the development of generative models such as OpenAI and Google DeepMind, enabling realistic manipulation of facial images and videos. While this technology has beneficial applications in media and entertainment, it also raises serious concerns related to misinformation, identity theft, and cybersecurity.

Detecting deepfake images is therefore a critical problem in computer vision and digital forensics. However, recent studies have shown that deepfake detection models are vulnerable to adversarial attacks, where small perturbations can significantly reduce model performance.

This project proposes a robust deepfake detection system that integrates:

- Deep learning-based classification
- Adversarial robustness analysis
- Adversarial training
- Explainable AI (XAI)
- Model trust and interpretability

The aim is to design a reliable and interpretable deepfake detector suitable for real-world deployment.

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2. Objectives

The primary objectives of this project are:

1. To develop a deepfake detection model using convolutional neural networks.
2. To analyze model vulnerability against adversarial attacks.
3. To improve robustness using adversarial training.
4. To evaluate model performance using standard metrics.
5. To apply Explainable AI techniques to understand model decisions.
6. To analyze model attention stability under adversarial perturbations.

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3. Dataset Description

The dataset used in this project contains labeled images classified into:

- REAL images
- FAKE (deepfake) images

The dataset was divided into:

- Training set
- Testing set

To ensure computational efficiency and balanced learning:

- 2500 samples per class were selected for training.
- 1000 samples per class were selected for testing.

This balanced subset ensures that the model does not develop bias towards a specific class.

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4. Methodology

4.1 Data Preprocessing

The following preprocessing techniques were applied:

- Image resizing to 224 × 224
- Normalization
- Random horizontal flipping
- Random rotation

These transformations improve model generalization and robustness.

4.2 Model Architecture

A pretrained ResNet-18 model was used. This architecture was originally proposed by Kaiming He and has demonstrated strong performance in image classification tasks.

Transfer learning was applied by:

- Using pretrained weights
- Replacing the final classification layer
- Fine-tuning the network

This reduces training time and improves performance.

4.3 Training

The model was trained using:

- Cross-entropy loss
- Adam optimizer
- GPU acceleration when available

Performance was evaluated based on training accuracy and loss.

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5. Evaluation Metrics

The model was evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix
- ROC curve and AUC

These metrics provide a comprehensive understanding of classification performance.

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6. Adversarial Attack Analysis

Deepfake detection systems are vulnerable to adversarial perturbations. To analyze this, the following attacks were implemented:

6.1 FGSM (Fast Gradient Sign Method)

Introduced by Ian Goodfellow, FGSM generates adversarial samples by adding perturbations in the direction of the gradient.

This attack showed that:

- The baseline model is highly sensitive to small perturbations.
- Model confidence drops significantly.

6.2 Blur Attack

Blur reduces high-frequency features, which deepfake detectors often rely on.

Observation:

- The model's performance dropped when blur was applied.

This indicates reliance on superficial texture.

6.3 Frequency Attack

Deepfake images contain high-frequency artifacts. Frequency domain manipulation was used to remove these features.

Observation:

- The baseline model showed reduced confidence.
- This confirms that the model relies on frequency artifacts.

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7. Adversarial Training

To improve robustness, adversarial training was implemented.

The model was trained using:

- Original images
- FGSM adversarial samples

This approach forces the model to learn more stable and meaningful features.

Results showed:

- Increased robustness
- Reduced confidence drop under attack
- Improved generalization

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8. Explainable AI (XAI)

Explainable AI was applied to understand the model's decision-making process. This approach is widely used in modern research institutions such as Stanford University and MIT.

8.1 Grad-CAM

Grad-CAM visualizes important image regions contributing to model predictions.

Findings:

- The baseline model focused on texture and artifacts.
- Attention shifted when adversarial perturbations were introduced.

8.2 Score-CAM

Score-CAM provided more stable and visually interpretable explanations.

This helped in:

- Understanding feature importance
- Comparing real and fake attention patterns

8.3 Correct vs Wrong Predictions

XAI analysis revealed that:

- Correct predictions focused on facial regions.
- Incorrect predictions focused on background and noise.

This insight improves model trust.

8.4 Attention Stability

Attention stability was measured before and after adversarial training.

Results showed:

- Reduced attention shift
- More consistent focus on semantic facial features.

This demonstrates robustness improvement.

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9. Experimental Results

9.1 Performance Improvement

After adversarial training:

- Accuracy improved
- Precision and recall improved
- F1-score increased

9.2 Robustness Improvement

Confidence under adversarial attack:

Before training:

- Fake confidence dropped significantly.

After training:

- Confidence remained stable.

This confirms that adversarial training enhances reliability.

9.3 ROC Analysis

ROC and AUC demonstrated strong classification ability.

The model showed high separability between real and fake images.

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10. Key Contributions

The main contributions of this work include:

1. Integration of deepfake detection with adversarial robustness.
2. Application of multiple attack methods.
3. Adversarial training for improved security.

4. Comprehensive evaluation.
5. Explainable AI for transparency.
6. Novel attention stability metric.
7. Frequency-based robustness analysis.

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11. Conclusion

This project demonstrates that deepfake detection models are vulnerable to adversarial and frequency-based perturbations. However, adversarial training significantly improves model robustness and stability.

Explainable AI techniques revealed that baseline models rely on superficial artifacts, while adversarial training encourages learning of meaningful semantic features.

The proposed framework enhances both reliability and trust, making it suitable for real-world applications.

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12. Future Work

Future extensions may include:

- Video deepfake detection
- Transformer-based models
- Multimodal deepfake detection
- Real-time detection systems
- Defense against adaptive adversarial attacks
- Deployment in cybersecurity pipelines

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13. Applications

- Digital forensics
- Media verification
- Cybersecurity
- Fake news detection
- Identity fraud prevention